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Course: Calculus Beta

Assignment: MyLab Math - Uge 4

Find the two square roots of $169i$.

If $z = re^{i\theta}$, then the expression $\sqrt[n]{re^{i\theta}}$ refers to the following collection of n distinct numbers. Note that $\sqrt[n]{r}$ refers to the single real n th root of r .

$$\sqrt[n]{re^{i\theta}} = \sqrt[n]{r} \exp \left[i \left(\frac{\theta}{n} + k \frac{2\pi}{n} \right) \right], k = 0, 1, 2, \dots, n-1$$

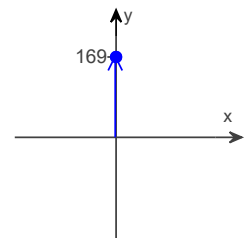
Since the goal in this case is to find the square roots of a number, $n = 2$.

The expression for $\sqrt[n]{re^{i\theta}}$ involves $\frac{2\pi}{n}$. Since $n = 2$, $\frac{2\pi}{n} = \frac{2\pi}{2} = \pi$.

This means the square roots of $z = re^{i\theta}$, w_0 and w_1 , are given by the following formula.

$$w_k = \sqrt{r} \exp \left[i \left(\frac{\theta}{2} + k\pi \right) \right], k = 0, 1$$

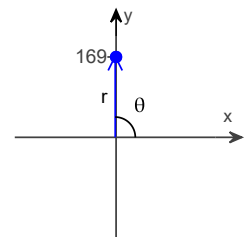
Plot $169i$ in the complex plane and determine its polar representation $re^{i\theta}$. This figure shows $169i$ in the complex plane.



Use this figure to find the values of r and θ , with $r \geq 0$ and $-\pi < \theta \leq \pi$, for the number $169i$.

$$r = 169$$

$$\theta = \frac{\pi}{2}$$



This means $169i = 169e^{\frac{i\pi}{2}}$.

Substitute $r = 169$ and $\theta = \frac{\pi}{2}$ into the formula for the square roots of a number from earlier.

$$\begin{aligned} w_k &= \sqrt{r} \exp \left[i \left(\frac{\theta}{2} + k\pi \right) \right] \\ &= \sqrt{169} \exp \left[i \left(\frac{\pi}{2} + k\pi \right) \right] && \text{Substitute } r = 169 \text{ and } \theta = \frac{\pi}{2}. \\ &= 13 \exp \left[i \left(\frac{\pi}{4} + k\pi \right) \right] && \text{Simplify.} \end{aligned}$$

Let $k = 0$ in the formula for w_k to find an expression for one square root, w_0 .

$$\begin{aligned} w_k &= 13 \exp \left[i \left(\frac{\pi}{4} + k\pi \right) \right] \\ w_0 &= 13 \exp \left[i \left(\frac{\pi}{4} + (0)\pi \right) \right] && \text{Substitute } k = 0. \\ &= 13e^{\frac{i\pi}{4}} && \text{Simplify.} \end{aligned}$$

Euler's Formula states that $e^{i\theta} = \cos \theta + i \sin \theta$. Use this to rewrite the expression from the previous step in terms of sine and cosine.

$$w_0 = 13e^{\frac{i\pi}{4}}$$

$$= 13 \left[\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right]$$

Use Euler's Formula.

Evaluate $\cos \frac{\pi}{4}$ and $\sin \frac{\pi}{4}$.

$$w_0 = 13 \left[\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right]$$

$$= 13 \left[\frac{\sqrt{2}}{2} + i \left(\frac{\sqrt{2}}{2} \right) \right]$$

Evaluate.

This means one square root of $169i$ is $w_0 = 13 \left[\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} \right] = \frac{13\sqrt{2}}{2} + \frac{13i\sqrt{2}}{2}$.

Let $k = 1$ in the formula for w_k to find an expression for the second square root, w_1 .

$$w_k = 13 \exp \left[i \left(\frac{\pi}{4} + k\pi \right) \right]$$

$$w_1 = 13 \exp \left[i \left(\frac{\pi}{4} + (1)\pi \right) \right]$$

Substitute $k = 1$.

$$= 13e^{\frac{5i\pi}{4}}$$

Simplify.

Use Euler's Formula to rewrite this in terms of sine and cosine.

$$w_1 = 13e^{\frac{5i\pi}{4}}$$

$$= 13 \left[\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right]$$

Use Euler's Formula.

Evaluate $\cos \frac{5\pi}{4}$ and $\sin \frac{5\pi}{4}$.

$$w_1 = 13 \left[\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right]$$

$$= 13 \left[-\frac{\sqrt{2}}{2} + i \left(-\frac{\sqrt{2}}{2} \right) \right]$$

Evaluate.

This means the second square root of $169i$ is $w_1 = 13 \left[-\frac{\sqrt{2}}{2} - \frac{i\sqrt{2}}{2} \right] = -\frac{13\sqrt{2}}{2} - \frac{13i\sqrt{2}}{2}$.

Thus, the two square roots of $169i$ are $\frac{13\sqrt{2}}{2} + \frac{13i\sqrt{2}}{2}$ and $-\frac{13\sqrt{2}}{2} - \frac{13i\sqrt{2}}{2}$.